Karachi Institute of Economics and Technology
College of Computing and Information Sciences
FINAL YEAR PROJECT – Fall-2026
PROPOSAL

SEMESTER	FALL 2026	YEAR	2026-2027
Degree Program		BSCS	

TITLE OF PROPOSED PROJECT

LegalDocReader: AI-Powered Legal & Policy Document Analyzer with Bilingual RAG & Risk Detection

Project Category (choose one)	✓ Product based	☐ Research-based

SUPERVISOR INFORMATION (To be filled by the office)

Supervisor Name:		Organization/ Designation	
Contact No:		email:	

STUDENT(S) INFORMATION

S#	Student ID	Name	
1	15372	Ahsan Mehmood	
Contact No:	03312051880	email:	Ahsanmehmood0 034@gmail.com
Expertise	Python, JavaScript, SQLite		
2	15020	Muhammad Mustafa	
Contact No:	03102034953	email:	15020@kiet.edu.p k
Expertise	Flask, Django, JavaScript, AI		
3	12906	Muhammad Sameer Ghori	
Contact No:	03192948920	email:	sameerghori.pk@ gmail.com
Expertise	Python,UI/UX Design, HTML/CSS		
4	15350	Hashir Shafiq	
Contact No:	03202092811	email:	shafiqhashir40@gm ail.com
Expertise	Python, Web Development, AI		

PROJECT AREA/TOOLS

Tools Required:	Python, Flask / FastAPI, SQLite, ChromaDB / FAISS, PyMuPDF, EasyOCR, OpenAI/Gemini API, HTML/CSS/JavaScript, VS Code, GitHub
Area/Specialization:	Artificial Intelligence (AI), Natural Language Processing (NLP), Web Application Development

SUMMARY OF PROPOSED PROJECT (MAXIMUM 400 WORDS)

LegalDocReader is an AI-powered document analysis system designed to simplify complex legal contracts, agreements, and corporate policy documents. By utilizing Retrieval-Augmented Generation (RAG) and Optical Character Recognition (OCR), the system allows users to upload PDF documents and interact with them using a simple

Q&A chat interface in both English and Urdu. The platform automatically flags risky or unusual clauses (such as hidden charges and unfair termination rules) and provides simplified breakdowns alongside an interactive document viewer. LegalDocReader solves the real-world challenge of understanding long legal documents with speed, bilingual clarity, and high accuracy without AI hallucination.

PROJECT OBJECTIVE(S)/OUTCOMES

- **Bilingual Accessibility:** Provide simple query and analysis capabilities in English and Urdu for complex documents.
- **Automated Risk Flagging:** Automatically detect, highlight, and categorize unfair penalties, unusual clauses, and hidden terms.
- **Hallucination-Free Q&A:** Ensure 100% grounded responses strictly derived from uploaded documents using RAG pipeline guardrails.
- **Simplified User Experience:** Deliver a side-by-side interactive viewer with risk tags and simplified summaries for quick evaluation.

SIMILAR APPLICATIONS (SIMILARITY/DISSIMILARITY)

S#	Name of App	Link (URL/Play Store)	Similar Features	New Features
1	ChatPDF	https://www.chatpdf.com	PDF Document Chat, Basic Text Summarization	Bilingual Urdu + English RAG Support, Risk/Unusual Clause Detection Engine
2	PDFgear AI	https://www.pdfgear.com	PDF Reader, Annotation Tools, Generic AI Chat	Contract Risk Flagging Dashboard, Grounded Hallucination-Free Guardrails
3	Spellbook	https://spellbook.com/	AI-powered Microsoft Word add-in that automatically reviews contracts, flags risky terms, and drafts custom legal clauses in real time.	Bilingual (English & Urdu) support tailored to local legal norms and contract terms.

Sustainable Development Goals (SDG)

1. ☐ No Poverty
2. ☐ Zero Hunger
3. ☐ Good Health and Well-being
4. ☐ Quality Education
5. ☐ Gender Equality
6. ☐ Clean Water and Sanitation
7. ☐ Affordable and Clean Energy
8. ☐ Decent Work and Economic Growth
9. ☐ Industry, Innovation, and Infrastructure
10. ☐ Reduced Inequalities
11. ☐ Sustainable Cities and Communities
12. ☐ Responsible Consumption and Production
13. ☐ Climate Action
14. ☐ Life Below Water
15. ☐ Life on Land
16. ☐ Peace, Justice, and Strong Institutions
17. ☐ Partnerships for the Goals

FYP Report at the time of FYP1 declaration (Soft Copy)

- **Chapter 1 : Introduction**
- User Authentication: Simple Login/Signup system using Email and Password.
- Document Upload & Management: PDF file upload, storage, and text extraction pipeline.
- Bilingual OCR Integration: Extracting text from scanned English and Urdu document images.
- **Chapter 2 : Literature Review/Process Review**
- Document Chunking & Vector Store: Processing document text into chunks and storing in a lightweight Vector DB (ChromaDB/FAISS).
- **Chapter 3 : Analysis Diagrams**

For FYP 2:

- Core RAG Chat Interface: Asking questions regarding the uploaded document and receiving context-grounded answers in Urdu/English.
- Basic Risk Flagging: Identifying key contractual terms (Termination, Payment, Liability).
- Summary Dashboard: Generating quick executive summaries with risk severity scores.

FYP Report at the time of FYP2 declaration (Hard Copy)

- Complete FYP Report
- Panaflex (2x5 ft.)

Additional Features by Jury Members:

For Proposal Defense Purpose

PROPOSED ADVISORY COMMITTEE		
S#	**Faculty Name**	**Signature**
1		
2		
3		
4		

FYP COMMITTEE			
S#	**Member(s) Name**	**Designation**	**Signature**
1	**Mr. Usman Khan**	Head, Final Year Projects	
2	**Dr. Maaz Bin Ahmed/ Dr. Sohail Imran/ Dr. Muhammad Fahad/ Dr. Ayub Latif**	Head of Dept.	
3	**Prof Dr Muhammad Khalid Khan**	Dean CIS	
	Date	09/05/2026	